

Problem A: Recursive Descent Parser

Subject

Description

Write a program that reads a source string from standard input, validates it against the provided token definitions and grammar rules, and generates the required output.

- If the input conforms to the grammar, **output each token along with its type, separated by a single space, and terminate each pair with a newline character.**
- If the input does not conform to the grammar, output only **"Invalid input"** with a **newline.**

NOTICE

- The test case input will **NOT** contain whitespace characters.
- Each test case input will consist of **only one line.**

The purpose of this boss attack is to test your understanding of writing a **recursive descent parser**. So, **it is forbidden to use lex, yacc, or any other parser generator tools**. Also, **"#include<regex>" is NOT allowed**. **Please write a program using the concept of a recursive descent parser in C++; otherwise, your score will be zero.**

Token Definition

Terminal	Regular Expression
IDENTIFIER	[a-zA-Z_]+
INTEGER	[0-9]+
STRING	"[a-zA-Z0-9_]+"
DATETIME	[0-9]{4}-[0-9]{2}-[0-9]{2}T[0-9]{2}:[0-9]{2}:[0-9]{2}
COLON	:
SEMI_COLON	;

NOTICE

Any string that matches the specified regex should be treated as valid. For example, **9999-99-99T99:66:77** will be recognized as a valid **DATETIME** token. **You don't need to worry about whether such values will actually appear in the test cases or whether they make sense as real dates.**

Grammar Rule

1. `program` → `entry SEMI_COLON`
2. `entry` → `record_log | record_val`
3. `record_log` → `DATETIME COLON STRING`
4. `record_val` → `IDENTIFIER COLON value`
5. `value` → `INTEGER | STRING`

Test Case

- The test case input will **NOT** contain whitespace characters.
- Each test case input will consist of **only one line**.

Input:

2023-10-27T10:30:00:"System_Booted";

Output:

2023-10-27T10:30:00 DATETIME

: COLON

"System_Booted" STRING

; SEMI_COLON

Input:

temperature:25;

Output:

temperature IDENTIFIER

: COLON

25 INTEGER

; SEMI_COLON

Input:

2023-10-27T10:30:00:"System_Booted"

Output:

Invalid input

Input:

user:"admin";

Output:

user IDENTIFIER

: COLON

```
"admin" STRING  
; SEMI_COLON
```